

AI FIT TRACK

AI-Based Personal Fitness Tracking and Recommendation System

PROJECT REPORT

A web-based fitness assistance system that calculates BMI and provides goal-based workout and basic diet recommendations using Python, Flask, HTML and CSS.

PROJECT TEAM

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Technology Stack

Python • Flask • HTML • CSS • Rule-Based AI Recommendation Logic

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1. Introduction

1.1 Project Overview

AI Fit Track is a web-based fitness assistance system designed to help users understand their basic fitness status and receive simple personalized recommendations. The application takes inputs such as name, age, weight, height, and fitness goal.

1.2 Problem Statement

Many users find it difficult to select suitable basic exercises and maintain a simple fitness routine. AI Fit Track provides a single interface where users can enter their details and receive recommendations according to their selected goal.

1.3 Need for the System

- Provides quick fitness-related calculations.
- Reduces the need for manual BMI calculation.
- Gives goal-based workout suggestions.
- Provides basic diet suggestions.
- Creates an easy-to-use interface for students and general users.

1.4 Scope

The prototype focuses on BMI calculation, fitness status classification, workout suggestions, and basic diet recommendations. It can later be extended with databases, machine learning, wearable-device integration, progress charts, and user accounts.

2. Objectives and Features

2.1 Main Objectives

- Develop a simple AI-assisted fitness recommendation application.
- Calculate Body Mass Index from height and weight.
- Classify the user's BMI into a basic category.
- Recommend exercises according to the selected fitness goal.
- Provide basic food and lifestyle suggestions.
- Create a clean and accessible web interface.

2.2 Major Features

- User input form.
- BMI calculation.
- BMI status identification.
- Weight-loss workout recommendations.
- Muscle-gain workout recommendations.
- General-fitness recommendations.
- Basic diet recommendations.
- Result page showing the personalized plan.

2.3 Target Users

The system can be used as an educational prototype for students and as a basic fitness planning interface for users who want simple recommendations. It is not intended to replace professional medical, nutritional, or fitness advice.

3. System Requirements and Technology

3.1 Hardware Requirements

- Computer or laptop.
- Minimum 4 GB RAM recommended.
- At least 500 MB free storage for the project environment.
- Internet connection is useful for installing packages and uploading the project to GitHub.

3.2 Software Requirements

- Windows, Linux, or macOS.
- Python 3.x.
- Visual Studio Code or another Python-compatible editor.
- Web browser such as Chrome, Edge, or Firefox.
- Git and GitHub for version control and project hosting.

3.3 Technology Stack

Python is used for the backend logic. Flask provides web routing and connects the input form to the recommendation functions. HTML creates the page structure, CSS provides the visual design, and the recommendation logic maps user goals to suitable predefined suggestions.

3.4 Why Flask?

Flask is lightweight and easy to understand for a college project. It allows the developer to create routes, receive form data, process it in Python, and return HTML templates.

4. System Modules and Architecture

4.1 User Input Module

The user enters name, age, weight, height, and fitness goal. The form validates required fields before sending the information to the backend.

4.2 BMI Calculation Module

The system converts height from centimetres to metres and calculates BMI using: $BMI = \text{weight (kg)} / \text{height}^2 \text{ (m}^2\text{)}$. The result is rounded for display.

4.3 Fitness Classification Module

The prototype uses basic BMI ranges to display Underweight, Normal, Overweight, or Obese. These categories are used only as a simple educational classification.

4.4 Recommendation Module

The recommendation engine checks the selected goal. For weight loss it provides activities such as walking, jogging, squats and planks. For muscle gain it provides strength-oriented bodyweight activities. General fitness receives balanced activities.

4.5 Diet Recommendation Module

The system provides simple food suggestions according to the selected goal. These are general examples and are not individualized medical nutrition plans.

4.6 Overall Flow

User → Input Form → Flask Backend → BMI Calculation → Goal Analysis → Recommendation Engine → Result Page.

5. Working Process and Algorithm

5.1 Step-by-Step Working

- Open the AI Fit Track home page.
- Enter the user's name and age.
- Enter weight in kilograms and height in centimetres.
- Select a fitness goal.
- Submit the form.
- Flask receives the form data.
- The backend calculates BMI.
- The BMI status is identified.
- The recommendation functions select workout and diet suggestions.
- The result page displays the personalized output.

5.2 Basic Algorithm

Step 1: Start. Step 2: Read user details. Step 3: Convert height to metres. Step 4: Calculate BMI. Step 5: Determine BMI category. Step 6: Read fitness goal. Step 7: Select workout plan. Step 8: Select diet suggestions. Step 9: Display result. Step 10: Stop.

5.3 Example

If a user enters weight = 60 kg and height = 165 cm, the system calculates BMI using $60 / (1.65 \times 1.65)$, giving approximately 22.04. The system then displays the corresponding basic BMI status and the recommendations associated with the selected goal.

6. Implementation and GitHub Structure

6.1 Project Folder Structure

- AI-Fit-Track/
- app.py
- requirements.txt
- templates/index.html
- templates/result.html
- static/style.css

6.2 Backend Implementation

The app.py file creates the Flask application, defines the home route, receives POST requests, calculates BMI, determines the status, and sends the recommendation lists to the result template.

6.3 Frontend Implementation

index.html contains the user form. result.html displays the calculated BMI, fitness status, workout plan, and diet suggestions. style.css controls layout, spacing, inputs, cards, and buttons.

6.4 Running the Project

Install dependencies using 'pip install -r requirements.txt' and start the application with 'python app.py'. The application can then be opened through the local Flask address shown in the terminal.

6.5 GitHub Upload Steps

- Create a new GitHub repository named AI-Fit-Track.
- Place all project files in the project folder.
- Add a README.md describing the project.
- Initialize Git in the project folder.
- Commit the files with a meaningful message.
- Push the project to the GitHub repository.

7. Advantages, Future Enhancements and Conclusion

7.1 Advantages

- Simple and easy-to-use interface.
- Fast BMI calculation.
- Goal-based recommendations.
- Lightweight Flask backend.
- Easy to demonstrate in a college project.
- Can be expanded into a larger fitness platform.

7.2 Future Enhancements

- User registration and login.
- SQLite/MySQL database.
- Personal progress history.
- Interactive charts and dashboards.
- Machine learning model for more personalized recommendations.
- Wearable or mobile-device integration.
- Calorie tracking and activity logging.
- Admin dashboard.
- PDF fitness report generation.

7.3 Conclusion

AI Fit Track demonstrates how Python, Flask, HTML, and simple AI-style recommendation logic can be combined to create a useful fitness application. The current version provides a strong college-project prototype and a foundation for future machine-learning and database features.

7.4 Note

Fitness and diet suggestions in this prototype are general educational recommendations. Users with medical conditions or specific nutritional needs should consult qualified healthcare or fitness professionals before following a fitness or diet plan.